

Volume & Surface Area — Full Solutions & Explanations

Digital SAT Style | Complete Working for Every Question

Q1 — EASY

■ **Correct Answer: C) 120 cm³**

STEP-BY-STEP SOLUTION:

1. $V = l \times w \times h = 4 \times 5 \times 6 = 120 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $15 = 4+5+6$, adding instead of multiplying.

✗ B: $60 = 4 \times 5 \times 3$, wrong dimension.

✗ D: $240 = 2 \times 120$, doubling.

■ **TRAP ALERT:** Adding dimensions instead of multiplying them.

■ **KEY INSIGHT:** Rectangular prism volume = length $\times$ width $\times$ height.

■ **FAST METHOD:** $4 \times 5 \times 6 = 120$.

Q2 — EASY

■ **Correct Answer: A) 5 cm**

STEP-BY-STEP SOLUTION:

1. $SA = 6s^2$. $6s^2 = 150 \rightarrow s^2 = 25 \rightarrow s = 5 \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: 6 is the number of faces, not the edge.

✗ C: $10 = 150/15$, wrong division.

✗ D: $25 = s^2$, not s .

■ **TRAP ALERT:** Reporting $s^2=25$ instead of $s=5$.

■ **KEY INSIGHT:** Cube SA = $6s^2$. Divide by 6 first, then take square root.

■ **FAST METHOD:** $s = \sqrt{(150/6)} = \sqrt{25} = 5$.

Q3 — EASY

■ **Correct Answer: C) 282.6 cm³**

STEP-BY-STEP SOLUTION:

1. $V = \pi r^2 h = 3.14 \times 9 \times 10 = 282.6 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $28.26 = \pi r^2$, area of one circular face only.

✗ B: $94.2 = 2\pi rh$, the lateral surface area.

✗ D: $942 = \pi(3^2)(10) \times \dots$ wrong.

■ **TRAP ALERT:** Using $\pi r \times h$ (missing r^2) or $\pi d^2 h$ (using diameter squared).

■ **KEY INSIGHT:** Cylinder volume = $\pi r^2 h$. Make sure you square the radius.

■ **FAST METHOD:** $\pi(9)(10) = 282.6$.

Q4 — EASY

■ **Correct Answer: B) 120 cm³**

STEP-BY-STEP SOLUTION:

1. $V = (1/3)(\text{base area})(\text{height}) = (1/3)(36)(10) = 120 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $60 = (1/6)(360)$, using wrong fraction.

✗ C: $180 = (1/2)(360)$, using 1/2 instead of 1/3.

✗ D: $360 = 36 \times 10$, forgetting the 1/3.

■ **TRAP ALERT: Forgetting the 1/3 factor for pyramids and cones.**

■ **KEY INSIGHT: Pyramid/cone volume = $(1/3)(\text{base area})(\text{height})$.**

■ **FAST METHOD: $(1/3)(36)(10) = 120$.**

Q5 — EASY

■ **Correct Answer: C) 113.04 cm³**

STEP-BY-STEP SOLUTION:

1. $V = (4/3)\pi r^3 = (4/3)(3.14)(27) = (4/3)(84.78) = 113.04 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $28.26 = \pi r^2$, the circle area, not sphere volume.

✗ B: $56.52 = 2\pi r^2$, surface area of a hemisphere.

✗ D: 904.32 uses $r=6$, doubling the radius.

■ **TRAP ALERT: Using $4\pi r^2$ (surface area formula) instead of $(4/3)\pi r^3$ (volume).**

■ **KEY INSIGHT: Sphere: $V=(4/3)\pi r^3$, $SA=4\pi r^2$. Don't mix them up.**

■ **FAST METHOD: $(4/3)\pi(27) = 36\pi \approx 113.04$.**

Q6 — MEDIUM

■ **Correct Answer: B) 8 cm**

STEP-BY-STEP SOLUTION:

1. Slant height, radius, and height form a right triangle.

2. $h^2 = \text{slant}^2 - r^2 = 100 - 36 = 64$.

3. $h = 8 \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $4 = 10 - 6$, subtracting instead of using Pythagorean.

✗ C: $12 = 10 + 2$, arbitrary.

✗ D: $16 = 2 \times 8$, doubling.

■ **TRAP ALERT: Computing $\text{slant} - r = 10 - 6 = 4$ instead of using the Pythagorean theorem.**

■ **KEY INSIGHT: In a cone: $\text{slant}^2 = r^2 + h^2$. This is a right triangle relationship.**

■ **FAST METHOD: $h = \sqrt{(100 - 36)} = \sqrt{64} = 8$. Recognize 6-8-10 triple.**

Q7 — MEDIUM

■ **Correct Answer: A) 5 cm**

STEP-BY-STEP SOLUTION:

1. $V = \pi r^2 h \rightarrow 500\pi = \pi r^2 (20)$.

2. $r^2 = 500/20 = 25 \rightarrow r = 5 \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: $10 = \sqrt{100}$, using wrong division.

✗ C: 15 from arithmetic error.

✗ D: $25 = r^2$, not r .

■ **TRAP ALERT:** Reporting $r^2=25$ instead of $r=5$.

■ **KEY INSIGHT:** To find radius from volume: isolate r^2 , then take square root.

■ **FAST METHOD:** $r = \sqrt{(500/20)} = \sqrt{25} = 5$.

Q8 — MEDIUM

■ **Correct Answer: C) 27**

STEP-BY-STEP SOLUTION:

1. Volume = $l \times w \times h$. If each is tripled: $V_{\text{new}} = 3l \times 3w \times 3h = 27(lwh) = 27V$.

2. Factor = $27 = 3^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 3 applies to 1D (length).

✗ B: $9 = 3^2$ applies to 2D (area).

✗ D: $81 = 3^4$, one factor too many.

■ **TRAP ALERT:** Using 3^2 (area scaling) instead of 3^3 (volume scaling).

■ **KEY INSIGHT:** Volume scales with k^3 where k is the linear scale factor. Area scales with k^2 .

■ **FAST METHOD:** Scale factor for volume = $3^3 = 27$.

Q9 — MEDIUM

■ **Correct Answer: B) 150.72 cm³**

STEP-BY-STEP SOLUTION:

1. $V = (1/3)\pi r^2 h = (1/3)(3.14)(16)(9) = (1/3)(452.16) = 150.72 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $48\pi \approx 150.8$, essentially the same — both correct forms.

✗ C: $452.16 = \pi r^2 h$, forgetting the $1/3$.

✗ D: $1356.48 = 3 \times 452.16$, multiplying by 3.

■ **TRAP ALERT:** Forgetting the $1/3$ factor for cones.

■ **KEY INSIGHT:** Cone volume = $(1/3)\pi r^2 h$. A cone is one-third of the cylinder with same base and height.

■ **FAST METHOD:** $(1/3)(3.14)(16)(9) = 150.72$.

Q10 — MEDIUM

■ **Correct Answer: C) 94 cm²**

STEP-BY-STEP SOLUTION:

1. $SA = 2(lw + lh + wh) = 2(12 + 15 + 20) = 2(47) = 94 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $47 = lw + lh + wh$, forgetting to multiply by 2.

✗ B: $60 = lwh$, that's the volume.

✗ D: $120 = 2 \times 60$, doubling the volume.

■ **TRAP ALERT:** Computing $lw+lh+wh=47$ and stopping without multiplying by 2.

■ **KEY INSIGHT:** SA of rectangular prism = $2(lw+lh+wh)$. Don't forget the factor of 2.

■ **FAST METHOD:** $2(12+15+20) = 2(47) = 94$.

Q11 — MEDIUM

■ **Correct Answer: C) $108\pi \text{ cm}^2$**

STEP-BY-STEP SOLUTION:

1. Curved surface = $2\pi r^2 = 2\pi(36) = 72\pi$.
2. Flat base = $\pi r^2 = 36\pi$.
3. Total SA = $72\pi + 36\pi = 108\pi \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 36π is just the base area.

✗ B: 72π is just the curved surface.

✗ D: $144\pi = 4\pi r^2$, full sphere SA without base.

■ **TRAP ALERT:** Forgetting to add the flat base area, or using the full sphere formula.

■ **KEY INSIGHT:** Hemisphere total SA = $2\pi r^2$ (curved) + πr^2 (base) = $3\pi r^2$.

■ **FAST METHOD:** $3\pi(36) = 108\pi$.

Q12 — MEDIUM

■ **Correct Answer: C) 8:27**

STEP-BY-STEP SOLUTION:

1. Linear ratio = 2:3.
2. Volume ratio = (linear ratio)³ = $2^3:3^3 = 8:27$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 2:3 is the linear (1D) ratio.

✗ B: $4:9 = 2^2:3^2$ is the area (2D) ratio.

✗ D: $16:81 = 2^4:3^4$, one power too many.

■ **TRAP ALERT:** Using $2^2:3^2=4:9$ (area ratio) instead of $2^3:3^3=8:27$ (volume ratio).

■ **KEY INSIGHT:** Linear $\rightarrow k$. Area $\rightarrow k^2$. Volume $\rightarrow k^3$.

■ **FAST METHOD:** Volume ratio = $(2/3)^3 = 8/27$.

Q13 — MEDIUM

■ **Correct Answer: B) $120\pi \text{ cm}^2$**

STEP-BY-STEP SOLUTION:

1. Lateral SA = $2\pi rh = 2\pi(5)(12) = 120\pi \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $60\pi = \pi rh$, forgetting the factor of 2.

✗ C: $170\pi = 120\pi + 50\pi$, adds a base area.

✗ D: $240\pi = 2 \times 120\pi$, doubling.

■ **TRAP ALERT:** Using πrh instead of $2\pi rh$ for the lateral surface.

■ **KEY INSIGHT:** Lateral SA of cylinder = $2\pi rh$. Think of "unrolling" the cylinder into a rectangle.

■ **FAST METHOD:** $2\pi(5)(12) = 120\pi$.

Q14 — HARD

■ **Correct Answer:** B) 5 cm

STEP-BY-STEP SOLUTION:

1. $\blacksquare^2 = r^2 + h^2 = 9 + 16 = 25$.

2. $\blacksquare = 5$ cm.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $4 = h$, the height, not slant height.

✗ C: $6 = r+h-1$, arbitrary.

✗ D: $7 = r+h$, adding.

■ **TRAP ALERT:** Confusing height with slant height. The slant height is ALWAYS longer than the height.

■ **KEY INSIGHT:** Cone right triangle: $\blacksquare^2 = r^2 + h^2$. 3-4-5 is a Pythagorean triple.

■ **FAST METHOD:** Recognize 3-4-5 triple $\rightarrow \blacksquare = 5$.

Q15 — HARD

■ **Correct Answer:** B) $500/3 \pi \text{ cm}^3$

STEP-BY-STEP SOLUTION:

1. Sphere inscribed in cube $\rightarrow$ diameter = side = 10, so radius = 5.

2. $V = (4/3)\pi(5^3) = (4/3)(125\pi) = 500\pi/3 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $166.7\pi \approx 500/3 \pi$, same answer differently written.

✗ C: $250\pi = (4/3)\pi(5^3) \times 1.5$, wrong factor.

✗ D: 1000π uses $r=10$ (diameter instead of radius).

■ **TRAP ALERT:** Using the cube side (10) as the radius. Radius = side/2 = 5.

■ **KEY INSIGHT:** Inscribed sphere: diameter = side of cube. Radius = side/2.

■ **FAST METHOD:** $r=5$. $V = (4/3)\pi(125) = 500\pi/3$.

Q16 — HARD

■ **Correct Answer:** B) 4 cm

STEP-BY-STEP SOLUTION:

1. Let r = radius. $h = 3r$.

2. $V = \pi r^2(3r) = 3\pi r^3 = 192\pi$.

3. $r^3 = 64 \rightarrow r = 4$ cm.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $r=3 \rightarrow V=81\pi \neq 192\pi$.

✗ C: $r=6 \rightarrow V=648\pi \neq 192\pi$.

✗ D: $r=8 \rightarrow V=1536\pi \neq 192\pi$.

■ **TRAP ALERT:** Using $h=3$ instead of $h=3r$.

■ **KEY INSIGHT:** Translate word relationships into algebra: "height is 3 times radius" $\rightarrow h=3r$.

■ **FAST METHOD:** $3r^3=192 \rightarrow r^3=64 \rightarrow r=4$.

Q17 — HARD

■ **Correct Answer:** A) 100 cm^3

STEP-BY-STEP SOLUTION:

1. Cylinder $V = \pi r^2 h = 300$.
2. Cone $V = (1/3)\pi r^2 h = (1/3)(300) = 100 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ B: $150 = 300/2$, using $1/2$ instead of $1/3$.
- ✗ C: $200 = 300 \times 2/3$, wrong fraction.
- ✗ D: $900 = 300 \times 3$, multiplying instead of dividing.

■ **TRAP ALERT:** Using $1/2$ instead of $1/3$. A cone is ONE-THIRD of its corresponding cylinder.

■ **KEY INSIGHT:** Cone = $(1/3) \times$ cylinder with same base and height.

■ **FAST METHOD:** $300/3 = 100$.

Q18 — HARD

■ **Correct Answer:** B) 96 m^3

STEP-BY-STEP SOLUTION:

1. Remaining depth = $5 - 3 = 2 \text{ m}$.
2. Volume needed = $8 \times 6 \times 2 = 96 \text{ m}^3$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $48 = 8 \times 6$, area of base only.
- ✗ C: $144 = 8 \times 6 \times 3$, volume already filled.
- ✗ D: $240 = 8 \times 6 \times 5$, total tank volume.

■ **TRAP ALERT:** Computing the total volume (240) or the filled volume (144) instead of the remaining.

■ **KEY INSIGHT:** Volume needed = base area $\times$ remaining height.

■ **FAST METHOD:** $8 \times 6 \times (5 - 3) = 8 \times 6 \times 2 = 96$.

Q19 — HARD

■ **Correct Answer:** D) 27

STEP-BY-STEP SOLUTION:

1. $V_{\text{large}} = (4/3)\pi(9^3) = (4/3)\pi(729) = 972\pi$.
2. $V_{\text{small}} = (4/3)\pi(3^3) = (4/3)\pi(27) = 36\pi$.
3. Number = $972\pi/36\pi = 27$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $3 = 9/3$, dividing radii instead of volumes.
- ✗ B: $9 = 3^2$, squaring the ratio.
- ✗ C: 18 from arithmetic error.

■ **TRAP ALERT:** Dividing radii ($9/3=3$) instead of volumes. Volume ratio = $(r_{\text{large}}/r_{\text{small}})^3$.

■ **KEY INSIGHT:** Number of small spheres = $V_{\text{large}}/V_{\text{small}} = (R/r)^3$.

■ **FAST METHOD:** $(9/3)^3 = 3^3 = 27$.

Q20 — HARD

■ **Correct Answer: D) 748 cm²**

STEP-BY-STEP SOLUTION:

1. Lateral SA = $2\pi rh = 2(22/7)(7)(10) = 440 \text{ cm}^2$.
2. Two bases = $2\pi r^2 = 2(22/7)(49) = 308 \text{ cm}^2$.
3. Total = $440 + 308 = 748 \text{ cm}^2$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 440 is only the lateral surface, missing both bases.
- ✗ B: $528 = 440 + 88$, adding only one base.
- ✗ C: 616 adds wrong base area.

■ **TRAP ALERT: Forgetting one or both circular bases in the total surface area.**

■ **KEY INSIGHT: Total SA = lateral + 2 bases = $2\pi rh + 2\pi r^2$.**

■ **FAST METHOD: $2\pi r(h+r) = 2(22/7)(7)(17) = 44 \times 17 = 748$.**

Q21 — HARD

■ **Correct Answer: D) $126\pi \text{ cm}^3$**

STEP-BY-STEP SOLUTION:

1. Original cone: $V = (1/3)\pi(6^2)(12) = 144\pi \text{ cm}^3$.
2. Cut at half height: top cone has $h=6$, $r=3$ (similar triangles).
3. $V_{\text{top}} = (1/3)\pi(3^2)(6) = 18\pi \text{ cm}^3$.
4. Frustum = $144\pi - 18\pi = 126\pi \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 42π is too small — uses wrong proportions.
- ✗ B: 63π would be frustum if top had $r=4.5$.
- ✗ C: 84π from arithmetic error.

■ **TRAP ALERT: The smaller cone's radius is NOT half the original height — use similar triangles to find it.**

■ **KEY INSIGHT: When cutting at half height, the smaller cone has half the radius (similar triangles). $V_{\text{small}} = (1/2)^3 = 1/8$ of original.**

■ **FAST METHOD: Original = 144π . Top cone = $(1/8)(144\pi) = 18\pi$. Frustum = 126π .**

Q22 — HARD

■ **Correct Answer: C) $4\sqrt{3} \text{ cm}$**

STEP-BY-STEP SOLUTION:

1. The circumscribed sphere passes through all 8 vertices of the cube.
2. The diameter of the sphere = space diagonal of cube = $\text{edge} \times \sqrt{3} = 8\sqrt{3}$.
3. Radius = $8\sqrt{3}/2 = 4\sqrt{3} \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $4 = \text{edge}/2$, that's the inscribed sphere radius.
- ✗ B: $4\sqrt{2} = \text{half the face diagonal}$.
- ✗ D: $8 = \text{the edge itself}$.

■ **TRAP ALERT:** Confusing inscribed (touching faces, $r = \text{edge}/2$) with circumscribed (through vertices, $R = \text{edge}\sqrt{3}/2$).

■ **KEY INSIGHT:** Circumscribed sphere radius = $(\text{edge} \times \sqrt{3})/2$. Inscribed sphere radius = $\text{edge}/2$.

■ **FAST METHOD:** $R = 8\sqrt{3}/2 = 4\sqrt{3}$.

Q23 — HARD

■ **Correct Answer:** C) 180 cm^3

STEP-BY-STEP SOLUTION:

1. Cross-section area = $(1/2)(6)(4) = 12 \text{ cm}^2$.
2. Volume = cross-section $\times$ length = $12 \times 15 = 180 \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $90 = 6 \times 15$, missing height.
- ✗ B: $120 = (1/2)(6)(4)(10)$, wrong length.
- ✗ D: $360 = 6 \times 4 \times 15$, forgetting the $(1/2)$.

■ **TRAP ALERT:** Forgetting the $(1/2)$ in the triangular cross-section area.

■ **KEY INSIGHT:** Prism volume = cross-section area $\times$ length.

■ **FAST METHOD:** $(1/2)(6)(4)(15) = 180$.

Q24 — HARD

■ **Correct Answer:** B) $320\pi \text{ cm}^3$

STEP-BY-STEP SOLUTION:

1. Volume = $\pi(R^2 - r^2)h = \pi(25 - 9)(20) = \pi(16)(20) = 320\pi \text{ cm}^3$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $160\pi = \pi(16)(10)$, wrong height.
- ✗ C: $480\pi = \pi(24)(20)$, wrong subtraction.
- ✗ D: $640\pi = \pi(32)(20)$, doubling the difference.

■ **TRAP ALERT:** Computing $\pi(R - r)^2 h = \pi(4)(20) = 16\pi$ instead of $\pi(R^2 - r^2)h$.

■ **KEY INSIGHT:** Hollow cylinder volume = $\pi(R^2 - r^2)h$. Subtract squared radii, not radii themselves.

■ **FAST METHOD:** $\pi(25 - 9)(20) = 320\pi$.

Q25 — HARD

■ **Correct Answer:** B) 15.28 cm

STEP-BY-STEP SOLUTION:

1. $V_{\text{cube}} = 12^3 = 1728 \text{ cm}^3$.
2. $V_{\text{cylinder}} = \pi r^2 h = 3.14(36)h = 113.04h$.
3. $113.04h = 1728 \rightarrow h = 1728/113.04 \approx 15.28 \text{ cm}$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $12 = \text{edge length}$, not the cylinder height.
- ✗ C: $24 = 2 \times 12$, doubling the edge.
- ✗ D: $48 = 1728/36$, forgetting π .

■ **TRAP ALERT:** Dividing by r^2 (36) instead of πr^2 (113.04).

■ **KEY INSIGHT:** When recasting: volume is conserved. Set $V_{\text{old}} = V_{\text{new}}$ and solve.

■ FAST METHOD: $h = 12^3/(\pi \times 36) = 1728/113.04 \approx 15.28$.

TOPIC SUMMARY — Volume & Surface Area

Common Mistakes (Top 3)

- Forgetting the $(1/3)$ factor for cones and pyramids.
- Confusing height with slant height in cone problems. $h^2 + r^2 = s^2$.
- Using linear scale factor instead of cubic: volume scales with k^3 , not k .

Must Remember (Top 3 Rules)

- Rectangular prism: $V = lwh$, $SA = 2(lw + lh + wh)$. Cylinder: $V = \pi r^2 h$, $SA = 2\pi r(r + h)$.
- Cone: $V = (1/3)\pi r^2 h$, $SA = \pi r(r + s)$. Sphere: $V = (4/3)\pi r^3$, $SA = 4\pi r^2$.
- The DSAT reference sheet provides volume formulas but NOT surface area formulas.

Expected on DSAT

- 2–3 questions per test. Common: cylinder volume, cone volume, scaling problems.
- Desmos: compute cube roots and square roots for solving equations quickly.